Q.1 ${\rm{The}}\,\,{\rm{domain}}\,\,{\rm{of}}\,\,\,{\rm{the}}\,\,{\rm{function }}f{\rm{(}}z{\rm{)}}\, = \,\frac{1}{{z\, - \,\left({1 + i} \right)}}\,\,{\rm{is}}\,\,\,___________________.$

A. $C\, \,\{ {\,1 + i\,} \}\,$

B. $C\, \,\{ {\,\,0\,} \}\,$

C. $C\, \,\{ {\,1 - i\,} \}\,$

D. $C\, \,\{ {\,\,1\,} \}\,$

Q.2 $$If\,\,we\,\,write\,\,the\,\,function\,\,w = {z^2} - z\,\,as\,\,w = u(x,y) + iv(x,y),\,\,then\,\,u(x,y) =$$

A. $$x{}^2 + {y^2}$$

B. $$x{}^2 - x - {y^2}$$

C. $$x{}^2 - {y^2}$$

D. ${{None}}\,\,{{ of}}\,\,{{ these}}$

Q.3 ${\rm{The}}\,\,{\rm{function }}\,f{\rm{(}}z{\rm{) }} = {\rm{ }}\frac{{\rm{1}}}{z}\,\,\,\,\,{\rm{is }}\,{\rm{__________}}\,{\rm{.}}$

A. not a polynomial

B. a polynomial

Q.4 ${\rm{If }}\,\,{\rm{the}}\,{\rm{ }}\,{\rm{5th}}\,\,{\rm{ roots }}\,\,{\rm{of }}\,\,{\rm{unity}}\,{\rm{ }}\,{\rm{are }}\,\,\,____________________.$

A. $5\,e{\,^{i\,\,({{{2k }}{5}})}},\,\,\,k\,\, = \,\,0,\,\,1,\,\,2,\,\,3$

B. $e{\,^{i\,\,({{{2k }}{5}})}},\,\,\,k\,\, = \,\,0,\,\,1,\,\,2,\,\,3,\,\,4$

C. $e{\,^{i\,\,({{{2k }}{5}})}},\,\,\,k\, = \,0,\,1,\,2,\,3$

D. $2\,e{\,^{i\,\,({{{2k }}{5}})}},\,\,\,k\,\, = \,\,0,\,\,1,\,\,2,\,\,3,\,\,4$

Q.5 Let z = 7 i, then Arg(z) = _____.

A. ${π 6}$

B. ${π 3}$

C. ${π 4}$

D. ${π 2}$

Q.6 ${\rm{The }}\,{\rm{values }}\,{\rm{of }}\,\,{\left({{\rm{2}}i} \right)^{\frac{1}{3}}}\,\,\,{\rm{are }}\,\,{\rm{__________}}{\rm{.}}$

A. $2{\,^{{1}{3}}} e{ ^{i\,\,({\,{{i\, }}{6} + {{2k }}{3}})}},\,\,\,\,\,\,k\, = \,0,\,1$

B. $2{\,^{{1}{3}}} e{ ^{i\,\,({\,{{i\, }}{6} + {{2k }}{3}})}},\,\,\,\,\,\,\,k\, = \,0,\,1,\,2$

Q.7 In the complex valued function w = f (z), the independent variable is _____.

A. z

B. w

Q.8 Domain of the function $$f\left(z \right) = \frac{1}{{z + \overline z }}$$ is _____.

A. $$z 2$$

B. $$z 1$$

C. $$z 0$$

D. $$z i$$

Q.9
${\rm{The}}\,\,{\rm{exponential}}\,\,{\rm{form}}\,\,{\rm{of}}\,\,{\rm{the}}\,\,{\rm{complex}}\,\,{\rm{number}}\,\,$
$= 2 - 2{\rm{ }}i{\rm{ }}\,\,{\rm{is _____}}{\rm{.}}$

A. $2\,e{\,^{i\,{ }{4}}}$

B. $2\,e{\,^{ - \,\,i\,{ }{4}}}$

C. $2\, 2 \,\,e{\,^{i\,{ }{4}}}$

D. $2\, 2 \,\,e{\,^{ - \,\,i\,{ }{4}}}$

Q.10 The punctured open disc with radius r and center z0 is represented by _____.

A. $| {z\,\, - \,\,{z_0}} |\,\,\, \,\,\,r$

B. $| {z\,\, - \,\,{z_0}} |\,\,\, = \,\,\,r$

C. $| {z\,\, - \,\,{z_0}} |\,\,\, > \,\,\,r$

D. $0\,\, < \,\,| {z\,\, - \,\,{z_0}} |\,\,\, < \,\,\,r$
